Sorathorn Thongpitukthavorn

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Education

Master of Engineering in Computer Science, *Cornell University (Cornell Tech)* 08/2026 – 05/2027 | New York, NY
Coursework: Applied Machine Learning, Optimization Methods, Networks & Markets, Product Studio.

Bachelor of Arts in Computer Science, *Boston University* 09/2022 – 05/2026 | Boston, MA
GPA: 3.61/4.00
Coursework: Distributed Systems, Cloud Computing, Database Systems, Analysis of Algorithms, Cybersecurity.

Work Experience

Software & Automation Engineer (Contractual), *Asia Inter Rice Co., Ltd.* 08/2025 – 08/2026 | Bangkok, Thailand
- Architected an automated purchase-order ETL platform (Python, Flask, Pandas, Electron) across 11 retail partners, standardizing PDF, XML, and Excel feeds into ERP records to cut processing time by >90%.
- Engineered a high-throughput C++ and Python multi-bank reconciliation engine backed by SQL databases, automating statement parsing, fuzzy transaction matching, and daily ledger auditing across accounts.
- Built an automated Google Cloud Vision OCR document pipeline to extract structured billing metrics from unstructured field images, eliminating 20+ hours/week of manual accounting entry.
- Shipped containerized full-stack billing and operations microservices (React, FastAPI, Docker) on GCP Linux VMs with blue-green deployments and automated LINE webhook workflows, reducing overdue receivables by 80%.

Software Developer Intern, *Datapro Computer Systems* 05/2025 – 08/2025 | Bangkok, Thailand
- Developed a full-stack identity-processing platform (React Native, TypeScript, Spring Boot, PostgreSQL) with real-time OCR document extraction, deployed via Docker on Linux servers.
- Architected automated Docker and GitLab CI/CD pipelines with strict staging-to-production parity, reducing deployment cycle times by 40% across engineering teams.
- Built automated end-to-end regression test suites using Selenium and Jira, resolving 30+ critical concurrency and performance bottlenecks prior to production release.

Technical Projects

Skail — Multi-Agent Autonomous Coding Harness , *Python, LangGraph, DeepAgents, SQLite, Git*
- Architected a concurrent multi-agent runtime orchestrating 3 specialist LLM agents across dependency-aware DAG workflows and standardized Model Context Protocol (MCP) tool servers.
- Engineered bounded context compaction and dynamic complexity-based model routing, cutting token overhead by 35% while maintaining deterministic crash recovery via transactional state journaling.

Project SCYLLA — Quantitative Options & Anomaly Terminal , *C++, Crow, Python, FastAPI, LightGBM*
- Built a hybrid market analytics engine pairing a multithreaded C++ Crow microservice with an asynchronous Python ingestion pipeline for real-time order-flow anomaly detection.
- Trained 5 LightGBM quantile models with native C-API inference and chronological walk-forward backtesting, backed by fault-tolerant REST streaming with circuit breakers and jittered retries.

HackTrackAI — Distributed Threat Intelligence & Cloud Streaming Platform , *Kubernetes, Apache Beam, AWS ECS*
- Engineered distributed Apache Beam streaming pipelines on Cloud Dataflow and Pub/Sub to process 10M+ telemetry records behind multi-zone GKE Layer 4 Load Balancers with <3s failover recovery.
- Integrated Mistral AI threat summarization with an interactive React and Deck.gl geospatial dashboard, containerizing microservices with automated CI/CD deployments to AWS ECS.

Technical Skills

- Languages: Python, C++, TypeScript, JavaScript, SQL, Java, Go, C, Bash.
- AI & Machine Learning: LangGraph, Model Context Protocol (MCP), Multi-Agent Orchestration, LiteLLM, LightGBM, PyTorch, Scikit-learn, OpenAI API, Cloud Vision OCR.
- Backend & Cloud Infrastructure: FastAPI, Spring Boot, Flask, React, React Native, Docker, Kubernetes (GKE), Apache Beam, Dataflow, Pub/Sub, AWS (ECS, EC2), GCP, Linux, NGINX.
- Databases & Tooling: PostgreSQL, Cloud SQL (MySQL), SQLite, Git, GitLab CI/CD, GitHub Actions, Selenium, Pandas, Electron.
- Certifications: CompTIA Security+, CompTIA Network+, CompTIA A+